CHAPTER

# System overview

IN THIS CHAPTER

## 3.1 What the unit does

The SalinBloc™ unit is built around HydroVolta's SonixED™ technology, which combines three mechanisms in one automated platform:

1. **Electrodialysis reversal (EDR):** an applied DC electric field draws dissolved ions — not water — through ion-exchange membranes from the diluate stream into the concentrate stream. Because water itself does not have to be forced through a membrane (unlike reverse osmosis), the process runs at much lower hydraulic pressure. Periodically reversing polarity cleans the membranes in place without chemicals and gives more stable performance as feed salinity varies.
2. **Integrated ultrasound antifouling:** ultrasound transducers built into the stack create acoustic cavitation at the membrane surface, disrupting scale nucleation and helping prevent silica fouling — this is the element of SonixED™ that HydroVolta holds a granted European patent on (2021).
3. **Custom membrane coating:** a proprietary coating gives the membranes monovalent-ion selectivity, so the process can target nitrate, sodium, chloride and fluoride removal while retaining calcium and magnesium in the product water.

The result is a saline feed stream split into a low-salinity product stream (*diluate*) and a concentrated reject stream (*concentrate*/brine), with chemical-free ultrasound antifouling keeping the membrane stack clean and extending electrode/membrane life (chapter 10). Typical applications: groundwater desalination and selective ion (e.g. nitrate) removal, industrial process-water reuse, brine valorization/concentration, and Zero Liquid Discharge (ZLD) support.


**Figure 3.1.** Suggested figure: exterior photograph of a delivered SalinBloc™ container (or container pair) on site, showing the overall footprint and connection points.

## 3.2 Field performance reference

HydroVolta publishes the following performance comparison from a 90-day industrial pilot (February–May 2023, cooling-tower blowdown at a chemical industry facility in Germany, feed TDS ≈4,500 ppm, silica 85 mg/l, high hardness), run head-to-head against reverse osmosis (RO) and a standard EDR system on the same feedwater (source: hydrovolta.com/technology.html and /projects.html). These figures describe that specific pilot and HydroVolta's general SonixED™ technology, not a performance guarantee for any individual SalinBloc™ unit; use them as context for what the technology is designed to achieve, and always verify against the site-specific commissioning data recorded in appendix A.

**Table 3.1.** Germany 2023 industrial pilot — SonixED™ vs. RO vs. standard EDR (source: hydrovolta.com)

| **Parameter** | **SonixED™** | **RO** | **Standard EDR** |
|---|---:|---:|---:|
| Water recovery | 90–98% | 50–75% | 70–85% |
| Energy consumption | 0.19–0.7 kWh/m$^3$ | 2.0–4.0 kWh/m$^3$ | 1.0–2.0 kWh/m$^3$ |
| Operating pressure | 0.5–2.5 bar | 25–35 bar | 1–2 bar |
| Chemical dosing | Minimal, no antiscalant | Required | Antiscalant needed |
| CIP frequency | 0.6×/month | 1×/month | 1×/month |
| Total treatment cost | \$0.30–0.40/m$^3$ | \$0.60–1.00/m$^3$ | \$0.40–0.80/m$^3$ |

HydroVolta states SonixED™'s general operating envelope as feed TDS 1,500–20,000 ppm ("strong fit" also for high silica >30 mg/l and high hardness >300 mg/l Ca+Mg feeds); seawater (>35,000 ppm) is stated as not the primary use case. Active and piloted SonixED™/SalinBloc™ deployments referenced by HydroVolta include a flagship groundwater desalination project in Morocco (seawater-intruded coastal aquifer, ≈7,000 ppm feed), a multi-site utility pilot with Water Corporation in Perth, Australia (selective nitrate and salinity removal), the Germany industrial pilot above, and earlier-stage projects in Brazil and Saudi Arabia.

## 3.3 Container layout

The system is delivered as a modular set of skids assembled inside one or more standard shipping containers (20 ft and 40 ft, High-Cube and Standard variants). Larger installations use **two** containers: a *pretreatment container* (filtration stages, chapter 8) and a *treatment container* housing the SonixED™ skid(s) (chapters 9 to 11 and 14), connected hydraulically. A documented 20 ft general-arrangement layout (drawing 2308CA1) shows internal dimensions of 6058 × 2438 × 2896 mm, housing a bank of SonixED™ stack racks, an electrical/control cabinet, a dosing/chemical section, three overhead storage tanks, and interconnecting piping; a companion top-view drawing (2308CA2) lays the skids out along the container floor: chemical skid, pump skid, tank skid, control 1 skid, filter skid, SonixED™ skid, control 2 skid.


**Figure 3.2.** Suggested figure: the skid top-view layout (based on drawing 2308CA2), redrawn/simplified for this manual, with each skid numbered and cross-referenced to its Part II chapter.

## 3.4 Process flow

1. Feed water enters the pretreatment container (2½" connection from source).

2. Pretreatment (disc filtration and ultrafiltration, chapter 8) conditions the water; filtered water leaves via a 2" connection.

3. Filtered water enters the treatment container (2" connection) and is split and pumped through parallel SonixED™ stacks (chapters 9 and 10), with three internal streams per stack: **diluate** (becomes the product/permeate), **concentrate** (becomes the brine/reject), and **electrolyte** (a separate loop feeding the electrode compartments, chapter 15).

4. Product (permeate) water exits the treatment container via a 2" outlet connection.

5. Reject/concentrate and pretreatment backwash/drainage are combined via a tee piece and routed to a drain tank via 3" drainage connections from both containers.

**NOTE**

TODO: a fully tagged P&ID with valve/instrument numbers exists for the BASF pilot (drawing SWHC-001/00, rev. 0.98) — see appendix B — but has not yet been confirmed as representative of the standard SalinBloc™ product configuration.

## 3.5 Main components

This section is a one-page index of every compartment covered in its own chapter later in this manual (Part II) — use it to jump straight to the part you need.

## 3.6 Control cabinet and HMI

The control system consists of a PLC with a touchscreen HMI, manual/override controls, and RS485 communication, housed in an IP55-rated cabinet. The skid ordering code allows two control-box configurations — **CB0** (PLC only) and **CB1** (PLC + touchscreen HMI) — and three cloud-connectivity tiers: **CL0** (local only), **CL1** (cloud monitoring), **CL2** (full cloud monitoring and control). See chapter 14 for details.

## 3.7 Nameplate and identification

TODO: no nameplate drawing, CE-marking template with filled-in fields, or Declaration of Conformity content was found (the `02_DoC's` and `05_SonixED stack` CE-marking folders in HydroVolta's engineering library are confirmed empty — see appendix C). The confirmed identification data is the datasheet labeling convention "Model: SONIXED-SKID, Version: 2025-A7" and the ordering code format `SONIXED-SKID-A-S-M-PR-V-CB-CL` (application type A0/A1/A2, stack count 1/2/3, measuring points 6/9/12, pressure sensors 6/9/12, valve count 2/4/6, control-box option CB0/CB1, cloud option CL0/CL1/CL2). Each skid physically measures approximately 3040 × 1340 × 2020 mm (l × b × h), on an aluminum-profile frame with a PVC/thermoplastic-spacer housing, mounted on lockable wheels for in-container mobility. HydroVolta's public SalinBloc™ brochure states container throughput by format: a **20 ft** unit handles 2–30 $m^3/h$ (≈44–660 $m^3$/day at 22 h/day operation); a **40 ft** unit handles 15–60 $m^3/h$ (≈330–1,320 $m^3$/day) — see section 3.2 and appendix A. No container weight figure was found in any source reviewed.


**Figure 3.3.** Suggested figure: photograph of the unit's nameplate once affixed, showing serial number, manufacture date, voltage/power rating, IP rating and CE mark.